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# Homogeneity of average annual air temperature time series for Croatia

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**ABSTRACT:** Homogeneity of the mean annual air temperature values at 2 m above ground for 22 stations in Croatia, in the period 1961–2000, is considered. The standard normal homogeneity test (SNHT) has been applied for this purpose, detecting both abrupt and gradual linear trend homogeneity breaks. For a group of lower-rank weather stations (poorer data quality), the results indicating inhomogeneity are sometimes difficult to explain. It has been shown that homogeneity breaks in multiannual averages are not higher than 0.5 °C, which could influence the natural trend analysis as well as some other statistical structures such as climatological normals, correlation functions or smoothed time series. Because of this, homogenization has usually to be adapted to the newest situation. On the other hand, the first principal component amplitude, describing more than 90% of the considered air temperature field variance, indicates that the spatial average of the mean annual temperature for the whole territory of Croatia is not dependent on inhomogeneities present in a particular time series. This result is in accordance with those previously obtained for Sweden and Switzerland as well as with some similar results obtained for larger continental and global scales. ['However, Easterling and Peterson (1995a, b) found that on very large spatial scales (half a continent to global), positive and negative homogeneity adjustments in individual station's maximum and minimum temperature time series largely balance out so when averaged into a single time series, the adjusted and unadjusted trend were similar.'–cited by Peterson *et al*. (1998)]. Copyright © 2009 Royal Meteorological Society

KEY WORDS homogenization; air temperature; Croatia

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## 1. Introduction

The time series of climatological variables (air temperature, humidity or pressure, wind speed and direction and others) are considered homogeneous if their variability is related only to regional weather and climate variability. However, observation of these variables is most frequently exposed to artificial influences caused by changes in the weather stations: their relocation, the replacement of observers and/or instruments, change in the environment, even including changes in observation rules. For example, during the nineteenth century, the screens for measuring air temperature were installed near windows, on the northern side of observation buildings, while in the twentieth century these screens were moved to an elevation of 2 m above ground and located on the observation sites (Begert *et al*., 2005). In addition, observation sites are continuously under the influence of environment change: the planting or uprooting of trees, the construction or redevelopment of houses, buildings, towers, roads, playgrounds and other infrastructural objects. This influence is stronger in big cities and the observation sites are frequently removed further to the periphery, thus 'running away' from urbanization. Of course, the reasons for changing the position of observation sites can be quite different in their nature. Artificial gradual change in the observation site environment causes an artificial gradual trend in air temperature, while observation site relocation usually causes a significant abrupt change in the air temperature average in comparison with the average of rather long periods before relocation.

Different statistical tests can be used for the detection of artificial changes or inhomogeneities in the statistical properties of climatological variables such as long-term averages, trends or standard deviations. They can be classified into two groups: *absolute* and *relative* homogeneity testing. Absolute homogeneity is considered within a particular time series, called the *candidate* time series, i.e. the time series which is tested. The disadvantage of these tests is the impossibility to make a distinction between inhomogeneity and a regional variation of weather and climate. On the other hand, when applying relative tests, the testing of the *candidate* time series is made in relation to the *reference* time series, which is assumed to exhibit the same natural variations of weather and climate and is assumed to be homogeneous. The reference series is used to separate the real climate signal and to highlight the artificial changes in the candidate series that do not occur in the reference series. A large number of both absolute and relative tests have been described by Marković (1975),

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Peterson *et al.* (1998) and Aguilar *et al.* (2003). Relative tests are more convenient for non-stationary time series with evidence of climatic changes. Because of the increasing number of non-stationary time series, relative tests have been recently applied more frequently than absolute tests. The most popular test was developed by Alexandersson (1986) to detect inhomogeneities in temperature and precipitation series: the standard normal homogeneity test (SNHT), which has its origin in Hawkins (1977). One version of this test indicates only abrupt breaks in homogeneity, while the second version indicates both abrupt and gradual changes of the mean value in the candidate series when compared with homogeneous reference series. All versions of SNHT give practical information: the date, size and significance of the break. This information is used in homogeneity testing and the adjusting of time series (Tuomenvirta and Alexandersson, 1996). Actually, Alexandersson's test can find only the most significant inhomogeneity in a time series. Therefore, the test has to be used in an iterative procedure. When a statistically significant inhomogeneity is found, the time series is homogenized for this inhomogeneity using the size of the inhomogeneity given by the test. Then, the time series is tested again and a second inhomogeneity may be found (Herzog and Müller-Westermeier, 1996). Elements for validation of the technique chosen can be also found in DeGaetano (1996), Vincent (1998), Vincent and Gullett (1999), Ducré-Robitaille *et al.* (2003), Li *et al.* (2004), Drogue *et al.* (2005), Brunetti *et al.* (2006), DeGaetano (2006) and Reeves *et al.* (2007).

The SNHT has been applied in Croatia by Likso (2003) for homogeneity testing of several annual air temperature time series for Croatia. [According to Marković (1975), average annual values can successfully be used in order to establish long-term modification. Variables averaged in shorter time units have greater mutual dependence among individual members and a larger variance which diminishes the test's discriminatory power.] Recently, the necessity has appeared for a more systematic approach to this topic because of the work on the Croatian Climate Atlas for the standard period 1961–1990. For this purpose, a more general version of the SNHT has been applied, which includes artificial linear trends in the temperature time series in addition to abrupt breaks in homogeneity (Section 2) and 22 time series of annual air temperature for the period 1961–2000 have been chosen (only the time series from conventional sensors have been considered because only short-time data were available from electronic sensors for the period considered), well covering the different climate regions of Croatia (Section 3). The results of homogeneity testing and the application of the homogenization procedure are described in Section 4. In addition, the differences between total linear trends for the homogenized and unhomogenized 22 average annual air temperature time series have been calculated and represented in Section 5. Consequently, it is possible to conclude that there is a gradual rise in the linear trend of mean annual temperature from the south to the north of the Croatian territory in the period considered, which is very probably the consequence of global warming.

## 2. Relative homogeneity testing methods for meteorological time series and homogenization

As it has been mentioned in Section 1, the Swedish meteorologist Alexandersson (1986) was the first to apply the SNHT for testing the homogeneity of meteorological variables. The test has been designed to remove the influence of the natural variation of weather and climate on a regional scale, i.e. (mostly) for the detection of artificially caused abrupt breaks in the homogeneity of meteorological time series.

Let $Y_i$ indicate the $i$th term (of total $n$) of the meteorological time series being tested ($Y$-series), while $X_{ji}$ is the $i$th term of the $j$th (of total $k$, also with $n$ terms each) reference time series for surrounding meteorological stations ($X$-series). The relative inhomogeneity to be detected as a series of $q_i$ ($q$-series) is defined as

$$q_i = \left(\frac{Y_i - \overline{Y}}{s}\right) - \left[\sum_{j=1}^{k} \rho_j{}^2 \left(\frac{X_{ji} - \overline{X}_j}{s_j}\right) \Big/ \sum_{j=1}^{k} \rho_j{}^2\right] \quad (1)$$

where $\overline{Y}$ and $\overline{X}_j$ are the sample averages for candidate and reference time series, respectively, $s$ is the standard deviation of the candidate, $s_j$ is the standard deviation for the $j$th reference series and $\rho_j$ is the correlation coefficient between the candidate and the $j$th reference series.

The SNHTs are applied to standardized time series

$$Z_i = \frac{q_i - \overline{q}}{s_q} \quad (2)$$

where $\overline{q}$ is the average value of the series of differences, $s_q$ is the corresponding standard deviation calculated by ($n$-1) weight in case $n$ is the number of the time series terms. It is assumed that $Z_i$ is $N(0,1)$ distributed. Alexandersson and Moberg (1997) discussed the problem of the assumption that standardization produces series with a unit variance (Equation (2)) and showed that standard deviations within two parts of a series, before and after a possible break, can be lower than the 'exact' value for the whole series. [Reeves *et al.* (2007) also discussed this problem with reference to Alexandersson (1986) only.] In addition to that, the same authors indicate that the use of the SNHT with two standard deviations can often give artificial breaks near the end of the series.

According to the *likelihood ratio* theory (e.g. Mood *et al.*, 1974; Lindgren, 1976; Wilks, 2006), the critical value of the test parameter can be found by:

$$T_0 = \max_{1<a<n}(T_a) = \max_{1<a<n}[a\overline{Z}_1 + (n-a)\overline{Z}_2] \quad (3)$$

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where

$$\overline{Z}_1 = \frac{1}{a}\sum_{i=1}^{a} Z_i$$

$$\overline{Z}_2 = \frac{1}{n-a}\sum_{i=a+1}^{n} Z_i \qquad (4)$$

Therefore, from an available standardized series of differences $Z_1, Z_2, \ldots, Z_n$ defined by Equation (2) and by the calculation of sub-sample means $\overline{Z}_1$ and $\overline{Z}_2$, the values of $T_a$ and $T_0$ are obtained (in short $T$-series). Several statements can be extracted:

1. If $T_0$ is greater, then the critical value candidate series should be considered to be inhomogeneous at a certain test level, e.g. with a probability of 95% (the probability of erroneous conclusion is 5% in this case).
2. $a$ indicates the term of the series after which a most probable inhomogeneity appears.
3. $\overline{q}_1$ and $\overline{q}_2$ are the mean values of the differences between the candidate and reference series before and after the possible break of homogeneity. Then, the difference $\overline{q}_2 - \overline{q}_1$ indicates the magnitude of the relative change in the candidate series compared to the reference series.

Now, a model can be introduced in which the 'mean level' of the series $q_1, q_2, \ldots q_n$ is changed gradually within its arbitrary length segment from $a$ to $b$. Then

$$\overline{Z}'_1 = \frac{1}{a}\sum_{i=1}^{a} Z_i \qquad (5)$$

and

$$\overline{Z}'_2 = \frac{1}{n-b}\sum_{i=b+1}^{n} Z_i \qquad (6)$$

indicate the arithmetic means in the series $Z_1, Z_2, \ldots, Z_n$, before and after the linear trend period, respectively. When $b = a + 1$, then the test parameter for gradual trends $T^t{}_0$ is reduced to the test parameter for abrupt breaks of homogeneity $T_0$. In addition to this, according to Alexandersson and Moberg (1997), the critical values for the test parameters $T_0$ and $T^t{}_0$, i.e. for abrupt breaks and gradual trends, respectively, are practically the same. It is usual practice to homogenize the time series so that their statistical parameters (e.g. mean values) for parts of the series before homogenization are adjusted to the statistical parameters (average values) of the parts after homogenization, i.e. the new state. Simply, the terms of the series before the break are corrected for the difference between the means after and before the break, respectively. If this difference is positive, the terms which are in chronological order before the break are increased for the difference and if it is negative, the same members are decreased for the same amount.

If a homogeneity break is in the form of linear trend, then the same correction is applied to the terms preceding the break, while for the terms which are under the influence of the linear trend the correction applied depends on the ordinal number of the term (see Alexandersson and Moberg, 1997).

What is the procedure in case of more than one break in the series $Z_1, Z_2, \ldots, Z_n$? In principle, break by break has to be solved in chronological order. After the first break has been detected, the described homogenization procedure can be applied. Then, the next break is detected and homogenized and so on, until the time series is completely homogenized. Therefore, a successive procedure should be applied to obtain the final homogenization of a time series, and this has been applied in this paper.

## 3. Time series of annual mean air temperature

For the Climate Atlas of Croatia, the homogeneity of the time series of mean annual temperature for 22 weather stations, indicated by number in Figure 1 and by name in Table II, almost uniformly distributed over Croatia, has been considered for the period 1961–2000. It is clear that the whole territory of the country is rather well covered, i.e. all climate zones are included: lowland, mountainous and coastal.

Visual inspection of the correlation matrix for the 'raw' time series represented in Table I(a) indicates that significant correlation coefficients have been obtained for any pair of weather stations and that there is no significant difference between the correlation coefficients of weather stations at similar distances. Although some authors, e.g. Peterson and Easterling (1994), suggest, in general, that the first difference series be used for the calculation of the correlation coefficients which serve as weights in Equation (1), this seems not to be obligatory in this case of relatively homogeneous time series. Only in the case of station number 2 (Karlovac) in Table I(a) the correlation coefficients are systematically several percentages lower

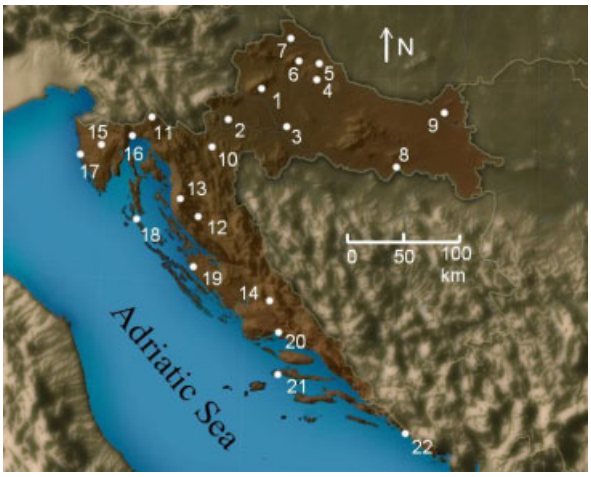



Figure 1. Geographical map of Croatia with weather station numbers. This figure is available in colour online at www.interscience.wiley.com/ijoc

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Table I. Correlation matrix of the (a) 'raw' annual average air temperature for Croatia in the period 1961–2000; (b) differences between the correlation coefficients for 'raw' and the first difference series of annual average air temperature for Croatia in the period 1961–2000.

(a)

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1.00 | **.91** | .96 | .96 | .98 | .96 | .97 | .94 | .94 | .96 | .89 | .95 | .86 | .82 | .89 | .89 | .88 | .90 | .90 | .79 | .74 | .66 |
| 2 | | **1.00** | **.91** | **.91** | **.91** | **.92** | **.94** | **.90** | **.95** | **.93** | **.86** | **.91** | **.84** | **.85** | **.80** | **.84** | **.78** | **.84** | **.78** | **.75** | **.74** | **.63** |
| 3 | | | 1.00 | .96 | .95 | .96 | .94 | .87 | .93 | .94 | .91 | .93 | .79 | .79 | .84 | .84 | .81 | .85 | .86 | .71 | .68 | .58 |
| 4 | | | | 1.00 | .96 | .95 | .95 | .88 | .94 | .94 | .87 | .94 | .82 | .79 | .88 | .86 | .82 | .88 | .89 | .75 | .71 | .62 |
| 5 | | | | | 1.00 | .95 | .97 | .93 | .94 | .96 | .87 | .94 | .83 | .80 | .87 | .85 | .85 | .88 | .87 | .76 | .71 | .63 |
| 6 | | | | | | 1.00 | .97 | .91 | .95 | .95 | .92 | .92 | .80 | .81 | .85 | .88 | .82 | .87 | .85 | .74 | .72 | .65 |
| 7 | | | | | | | 1.00 | .94 | .96 | .98 | .92 | .93 | .86 | .83 | .85 | .88 | .84 | .89 | .85 | .76 | .73 | .65 |
| 8 | | | | | | | | 1.00 | .93 | .93 | .83 | .92 | .82 | .83 | .80 | .85 | .89 | .86 | .84 | .79 | .77 | .72 |
| 9 | | | | | | | | | 1.00 | .94 | .89 | .94 | .84 | .86 | .84 | .87 | .82 | .88 | .84 | .78 | .76 | .67 |
| 10 | | | | | | | | | | 1.00 | .91 | .95 | .87 | .86 | .87 | .88 | .87 | .91 | .87 | .81 | .79 | .70 |
| 11 | | | | | | | | | | | 1.00 | .86 | .86 | .84 | .81 | .89 | .82 | .86 | .83 | .75 | .74 | .64 |
| 12 | | | | | | | | | | | | 1.00 | .83 | .88 | .87 | .89 | .88 | .92 | .90 | .83 | .80 | .71 |
| 13 | | | | | | | | | | | | | 1.00 | .88 | .87 | .92 | .88 | .92 | .86 | .86 | .84 | .74 |
| 14 | | | | | | | | | | | | | | 1.00 | .88 | .91 | .82 | .91 | .85 | .90 | .89 | .82 |
| 15 | | | | | | | | | | | | | | | 1.00 | .90 | .83 | .93 | .94 | .87 | .84 | .76 |
| 16 | | | | | | | | | | | | | | | | 1.00 | .92 | .96 | .91 | .90 | .89 | .80 |
| 17 | | | | | | | | | | | | | | | | | 1.00 | .91 | .89 | .85 | .84 | .75 |
| 18 | | | | | | | | | | | | | | | | | | 1.00 | .94 | .92 | .90 | .81 |
| 19 | | | | | | | | | | | | | | | | | | | 1.00 | .89 | .87 | .78 |
| 20 | | | | | | | | | | | | | | | | | | | | 1.00 | .97 | .89 |
| 21 | | | | | | | | | | | | | | | | | | | | | 1.00 | .93 |
| 22 | | | | | | | | | | | | | | | | | | | | | | 1.00 |

(b)

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | .00 | −.07 | −.01 | −.01 | −.01 | −.03 | −.01 | −.02 | −.01 | .00 | −.01 | .03 | .06 | −.01 | .04 | .06 | .03 | .09 | .05 | −.04 | −.05 | −.08 |
| 2 | | .00 | −.05 | −.05 | −.03 | −.05 | −.01 | −.05 | .00 | −.03 | .00 | −.01 | .04 | .01 | −.02 | .01 | −.06 | .02 | −.06 | −.09 | −.06 | −.08 |
| 3 | | | .00 | .01 | −.01 | .00 | .00 | −.06 | .00 | −.02 | .05 | −.01 | .06 | −.01 | .05 | .07 | −.02 | .07 | .05 | −.08 | −.08 | −.11 |
| 4 | | | | .00 | −.01 | −.03 | −.02 | −.07 | .00 | .00 | .00 | .04 | .07 | .00 | .06 | .05 | −.01 | .10 | .06 | −.05 | −.04 | −.09 |
| 5. | | | | | .00 | −.03 | .01 | −.01 | .00 | .02 | .01 | .05 | .09 | .02 | .06 | .07 | .03 | .10 | .05 | −.03 | −.04 | −.08 |
| 6. | | | | | | .00 | −.01 | −.05 | .00 | .00 | .03 | .00 | .02 | −.01 | .00 | .04 | −.04 | .05 | .00 | −.09 | −.08 | −.09 |
| 7. | | | | | | | .00 | .01 | .02 | .02 | .00 | .04 | .06 | .02 | .03 | .05 | .01 | .08 | .03 | −.04 | −.02 | −.05 |
| 8. | | | | | | | | .00 | −.03 | .02 | −.01 | .01 | .08 | .02 | −.01 | .08 | .04 | .08 | −.02 | −.02 | −.01 | .00 |
| 9. | | | | | | | | | .00 | .02 | .04 | .03 | .07 | .03 | .03 | .06 | −.01 | .07 | .01 | −.04 | −.03 | −.06 |
| 10 | | | | | | | | | | .00 | .01 | .03 | .07 | .02 | .04 | .08 | .03 | .09 | .05 | −.01 | .01 | .00 |
| 11 | | | | | | | | | | | .00 | .04 | −.02 | −.01 | −.05 | .02 | −.06 | .02 | .00 | −.08 | −.05 | −.07 |
| 12 | | | | | | | | | | | | .00 | .11 | −.01 | .04 | .07 | .03 | .09 | .03 | −.04 | −.04 | −.07 |
| 13 | | | | | | | | | | | | | .00 | .01 | −.01 | .02 | .02 | .04 | .03 | .00 | .02 | −.03 |
| 14 | | | | | | | | | | | | | | .00 | −.06 | −.03 | −.08 | −.03 | −.07 | −.04 | −.01 | −.04 |
| 15 | | | | | | | | | | | | | | | .00 | −.06 | −.11 | −.01 | −.01 | −.07 | −.08 | −.12 |
| 16 | | | | | | | | | | | | | | | | .00 | −.01 | .01 | .00 | −.03 | −.02 | −.07 |
| 17 | | | | | | | | | | | | | | | | | .00 | −.02 | −.05 | −.09 | −.09 | −.10 |
| 18 | | | | | | | | | | | | | | | | | | .00 | .01 | −.04 | −.04 | −.05 |
| 19 | | | | | | | | | | | | | | | | | | | .00 | −.06 | −.08 | −.11 |
| 20 | | | | | | | | | | | | | | | | | | | | .00 | −.01 | −.04 |
| 21 | | | | | | | | | | | | | | | | | | | | | .00 | −.01 |
| 22 | | | | | | | | | | | | | | | | | | | | | | .00 |

than those of its pairs with neighbouring stations (numbers 1 and 3) but that difference cannot significantly influence the weights in Equation (1) because the total sum of the coefficients is also lower. This correlation matrix also represents the first insight into data quality and their homogeneity. This presumption is justified: the matrix of differences between correlation coefficients for 'raw' and the first difference series, represented in Table I(b), indicates that those differences are not significant. They are mostly randomly distributed within a range from about −0.1 to 0.1, i.e. to about 10% of the correlation coefficients' range, which is, statistically, at a white-noise level for samples of 40 terms and, therefore, they cannot significantly influence the weights in Equation (1). Although this is formally true, some of the differences shown are very probably a consequence of time series inhomogeneity (see neighbouring pairs with station number 2), as it will be shown later in Section 5 of this paper, where correlation coefficients are compared for homogenized and unhomogenized smoothed time series.

In addition to the correlation functions relating to the weights in Equation (1), the statistical properties of the $q$-series are of great interest for the requirements of the homogenization technique to be met, i.e. Alexandersson's SNHT. First of all, it is presumed that the $q$-series has a normal distribution, a constant standard deviation and a non-significant serial correlation (autocorrelation) in the homogeneous parts of the series. These properties of the $q$-series have been examined after the homogenization procedure, which is described in Section 4. Thus, all $q$-series have the same sample size and the test statistics are comparable for all of them.

Visual inspection of Figure 2, where sample histograms are represented for weather station number 10 (Ogulin) together with the adjusted histograms of Gaussian (normal) distribution, indicates a reasonable fitting (see e.g. Wilks, 2006). Although this is not the case with all $q$-series, their fittings are within the statistically acceptable range at 5% significance level (Figure 3).

Serial correlation coefficients have been calculated for all $q$-series from 1 to 20 lags. From the results, which are shown in Figure 4, it is evident that the serial correlation function values are much lower than the values of the 5% significance level $t$-test statistic (see e.g. Pavlić, 1970).

Finally, standard deviations have been considered for the $q$-series. For this purpose, 20-year window standard deviations have been calculated for all $q$-series. It is evident from Figure 5, where the corresponding standard

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Table II. The results of homogeneity testing of the mean annual air temperature time series from 22 weather stations in Croatia, for the period 1961–2000.

| Weather station | | Homogeneity break | | Mean change for Y-series | | Potential cause of inhomogeneity |
|---|---|---|---|---|---|---|
| | $T_0^t$ | *a* | *b* | °C | °C/year | |
| 1. Zagreb-Maksimir | 11.8* | 1964 | 1966 | −0.20 | −0.10 | General urbanisation, especially construction of the *Kraš* factory and tennis courts in the vicinity, in the 80's and 90's of last century, respectively |
| | 20.4* | 1979 | 1993 | 0.24 | 0.02 | |
| | 10.0* | 1977 | 1993 | 0.11 | 0.01 | |
| | 9.9* | 1991 | | 0.11 | | |
| | 9.2* | 1995 | 1998 | −0.13 | −0.04 | |
| 2. Karlovac | 9.4* | 1979 | 1991 | 0.26 | 0.02 | Urbanisation and relocation of the weather station on 9 November 1992 |
| | 28.7* | 1992 | | −0.59 | | |
| | 7.8 | 1968 | | −0.15 | | |
| 3. Sisak | 17.4* | 1968 | 1970 | 0.28 | 0.14 | Urbanisation and change of environment |
| | 9.2* | 1975 | 1986 | −0.13 | −0.01 | |
| 4. Bjelovar | 13.2* | 1985 | 0.33 | 0.17 | | Change in environment |
| | 13.9* | 1988 | | −0.24 | | |
| 5. Durdevac | 12.3* | 1966 | 1971 | −0.24 | −0.05 | Change in environment |
| | 14.8* | 1977 | | 0.20 | | |
| | 11.2* | 1987 | | 0.15 | | |
| 6. Križevci | 12.8* | 1970 | −0.06 | −0.03 | | Fruit tree planting in the 80's of last century |
| | 24.1* | 1981 | 1983 | −0.21 | −0.11 | |
| 7. Varaždin | 13.6* | 1971 | | −0.25 | | Relocation of the weather station on 1 January 1972 |
| | 6.9 | 1982 | | 0.11 | | |
| | 11.6* | 1994 | 1997 | −0.21 | −0.07 | |
| 8. Slavonski Brod | 17.5* | 1969 | | −0.45 | | From 1 January 1970, data are from the new location (parallel work of two station for one year); urbanisation |
| | 18.7* | 1986 | 1992 | 0.39 | 0.07 | |
| | 16.6* | 1996 | | −0.36 | | |
| | 6.3 | 1969 | | −0.12 | | |
| 9. Osijek | 7.5 | 1968 | | −0.18 | | Relocation of the weather station, on 17 October 1991 |
| | 8.5* | 1990 | 1992 | −0.18 | −0.09 | |
| 10. Ogulin | 2.5 | 1968 | | | | Probably no break in homogeneity |
| 11. Parg | 13.8* | 1968 | | 0.41 | | Probable influence of the reconstruction of the station site? |
| | 20.9* | 1974 | 1985 | −0.48 | −0.04 | |
| 12. Gospić | 4.4 | 1976 | | 1979 | | Probably no break in homogeneity |
| 13. Zavižan | 3.2 | 1984 | | | | Probably no break in homogeneity |
| 14. Knin | 8.6* | 1964 | 1967 | 0.28 | 0.09 | Station relocated twice: 16 May 1971 and 22 December 1988; war in 1995 |
| | 9.7* | 1995 | | −0.25 | | |
| 15. Pazin | 19.2* | 1969 | 1982 | | 0.03 | Change in environment |
| | 18.6* | 1987 | 1989 | −0.30 | −0.15 | |
| 16. Rijeka | 14.6* | 1976 | | −0.18 | | Relocation of the weather station on 1 December 1977 |
| 17. Rovinj | 17.3* | 1979 | 1981 | −0.41 | −0.21 | Relocation of the weather station on 17 December 1979; change in environment |
| | 14.6* | 1986 | 1989 | 0.37 | 0.12 | |
| | 9.1* | 1987 | | 0.16 | | |
| 18. Mali Lošinj | 4.3 | 1970 | 1972 | | | Probably no break in homogeneity |
| 19. Zadar | 7.4 | 1963 | 1965 | 0.29 | 0.15 | Station relocated twice: 1 Jan. 1965 and 27 June 1995 |
| | 16.1* | 1994 | | 0.29 | | |
| 20. Split-Marjan | 14.5* | 1963 | 1969 | −0.22 | −0.04 | Change in environment |
| | 16.4* | 1979 | | 0.14 | | |
| 21. Hvar | 6.8 | 1964 | | | | Probably no break in homogeneity |
| 22. Dubrovnik | 10.3* | 1970 | | −0.24 | | Relocation of the weather station on 12 May 1979 |

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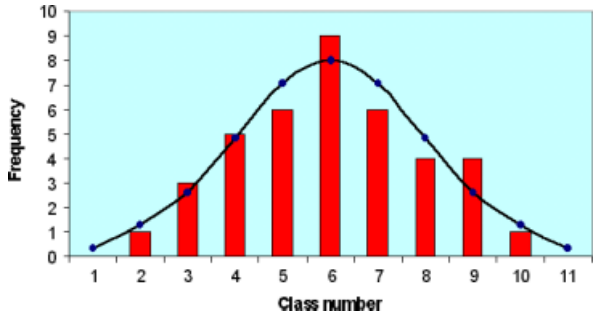



Figure 2. Histograms of $q$-series for weather station number 10 (rectangles) and Gaussian approximation (smooth curve). This figure is available in colour online at www.interscience.wiley.com/ijoc

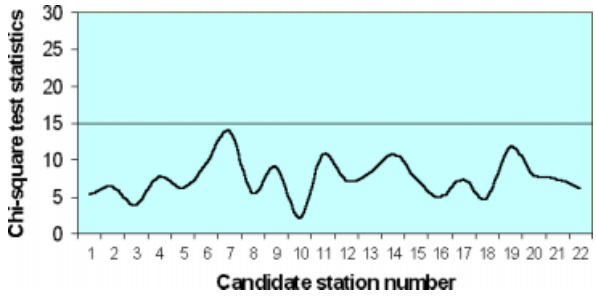



Figure 3. Sample goodness-of-fit statistic of 22 $q$-series for 22 weather stations in Croatia (lower thick curve) and the corresponding theoretical 5% significance level value of the theoretical Chi-square test statistic (upper thin line). This figure is available in colour online at www.interscience.wiley.com/ijoc

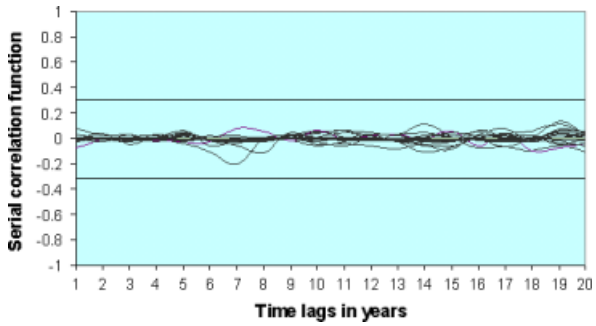



Figure 4. Serial correlation functions of $q$-series for 22 weather stations in Croatia (twisting lines) and the 5% significance level $t$-test statistics (straight lines). This figure is available in colour online at www.interscience.wiley.com/ijoc

deviations are shown for 22 $q$-series for Croatia, that there are no significant changes in standard deviations as the calculation window moves from the beginning to the end of the $q$-series.

## 4. The results of homogeneity testing and homogenization

Using the theory described in Section 2, the authors have produced algorithms and computing programmes for the calculation of the $T$- and $q$-series, $T^t{}_0$ test quantity, as well as the beginning and the end of the linear trends in the $q$-series. If the difference is only 1 year, i.e. $b = a +$

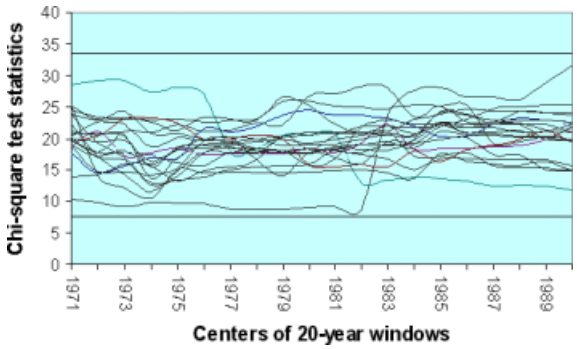



Figure 5. Twenty-year window standard deviations for 22 q-series in Croatia (twisting lines) and the 5-percent significance levels of Chi-square test statistic (straight lines). This figure is available in colour online at www.interscience.wiley.com/ijoc

1, then an abrupt break of homogeneity is the case. Thus, using a unique algorithm and graphical visualization of the $T$-series, abrupt and gradual breaks of homogeneity have been detected in the $q$-series of mean annual air temperature in Croatia. An algorithm and the necessary computing programmes have been developed for the homogenization of the time series. The programmes have been developed in MS-DOS FORTRAN 77 and can be run on personal computers. The results have been compared with those of the internationally accepted *SNHT for Windows* software which was developed by the Danish Meteorological Institute (DMI) for the detection of abrupt breaks and trends (Steffensen, 1996) and also used by Likso (2003).

As the density of weather stations is higher in the lowland than in the mountainous and coastal regions, reference series for the lowland part of Croatia have been constructed on the basis of five time series from the closest weather stations, while for the mountainous and coastal regions data from the three closest weather stations have been used. It turned out that the results with data from three weather stations are as stable as those from five weather stations. This preliminary conclusion is in agreement with the results by Moberg and Alexandersson (1997), which show that spatial average is 'immune' from independent inhomogeneities of a particular time series, which eventually mutually annul each other. This statement will be also confirmed in this study.

The numerical results of the homogeneity testing of 22 time series of 40-year mean annual values of air temperature are presented in Table II in the same order as in Figure 1. Those test quantities which exceed the critical value of $T^t{}_0 = 8.10$, by a confidence level of 95%, in the case of $n = 40$, are marked by an asterisk. [The critical value of $T^t{}_0 = 8.10$ has been taken from Alexandersson and Moberg (1997). Khaliq and Ouarda (2007) improved the accuracy of critical values and extended the sample sizes, but still, for the considered sample of 40 terms, the result is practically the same as the one by Alexandersson and Moberg (1997).] Values lower than 8.10, but still close to it, are sometimes considered to be a signal of homogeneity break, while others, i.e. low enough at

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corresponding locations in the time series, are not considered to be breaks in homogeneity. Further, in Table II, $a$ indicates the year preceding a homogeneity break and $b$ is the last year within the linear trend period. In the case of an abrupt break of homogeneity, i.e. when $b = a + 1$, then $b$ is omitted.

The differences between the average values of the $Y$-series for the parts preceding the homogeneity breaks and those following the breaks, as well as the 'linear trend' per year, for cases with such a trend, are represented in the fourth column of Table II. Finally, in the last column, possible causes of homogeneity breaks are described: relocation of observation site and change in the site environment (e.g. urbanization in the case of big cities). Other possible causes of inhomogeneity such as change of instrumentation or observer, including possible isolated natural local influences, are not considered here.

From Table II, it is obvious that a certain number of homogeneity breaks coincide with the times of weather station relocation: Karlovac (1992), Varaždin (1971), Slavonski Brod (1969), Osijek, Rijeka and Zadar (1994). In some cases, i.e. in Rovinj, Zadar (1965) and Dubrovnik, relocation of weather stations took place, but significant homogeneity breaks are not indicated.

A diagnosis of the graphical representation of the $T$- and $q$-series additionally explains the results represented in Table II. The simplest representations are for the mentioned five weather stations (of a total of 22) for which no inhomogeneity has been detected. A very illustrative example is Karlovac (Figures 6(a) and 7(a)), where both considered types of inhomogeneity have been detected and successfully explained. On the other hand, in some

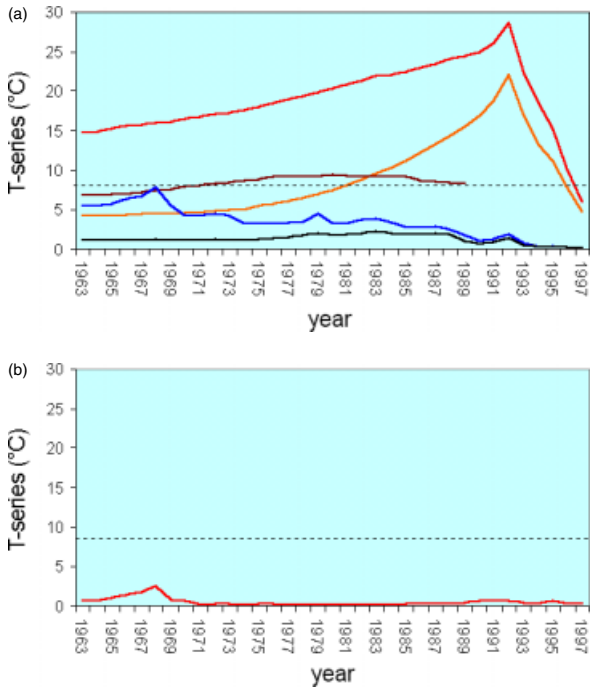



Figure 6. Representation of $T$-series for (a) Karlovac and (b) Ogulin. This figure is available in colour online at www.interscience.wiley.com/ijoc

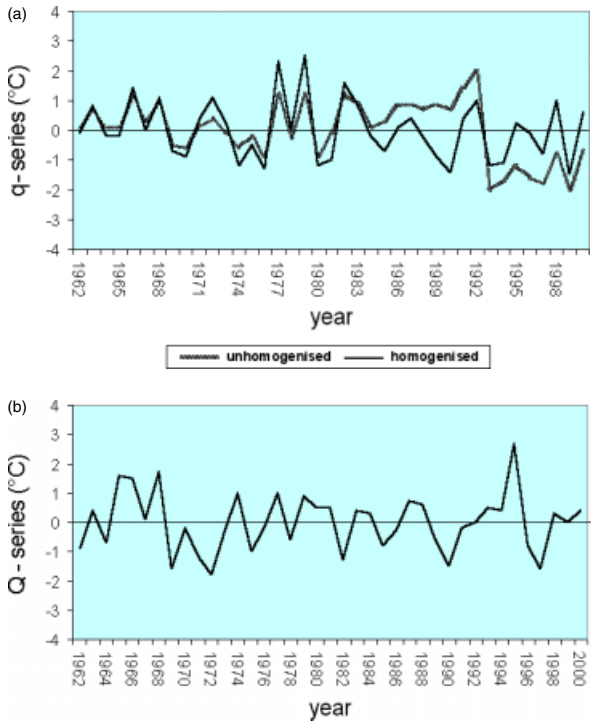



Figure 7. Representation of $q$-series for (a) Karlovac before and after homogenization and (b) Ogulin. This figure is available in colour online at www.interscience.wiley.com/ijoc

cases (Bjelovar, Durdevac, Parg, Pazin, Rovinj and some others) the explanation is not very clear even after the homogenization of the time series. It seems that the air temperature time series for, in data quality, lower-rank weather stations are more confusing, when it comes to homogeneity, than the air temperature time series for higher-rank stations. The latter are frequently under a stronger influence of urbanization because they are usually connected with big urban centres. (In meteorological practice, conventional weather stations are grouped in categories: main or principal, ordinary, rainfall etc. Usually, the best quality data come from main stations with professional observers. As this is not always the case, weather stations can be ranked according to data quality, as has been done here.) For those which are not affected by urbanization, no inhomogeneity has been established (e.g. Ogulin, Gospić and Zavižan). The $T$- and $q$-series for the last group of weather stations are represented in Figures 6(b) and 7(b), respectively.

On the other hand, at the Zagreb-Maksimir, Karlovac (1968), Bjelovar, Durdevac, Varaždin (1982), Slavonski Brod (1969 and 1996), Parg, Knin, Rovinj, Split-Marjan and Dubrovnik weather stations, abrupt breaks of homogeneity have been discovered although relocation of these weather stations has not been registered. Possible reasons could be the building of infrastructural objects like the tennis court near the Zagreb-Maksimir weather station towards the end of last century. At some weather stations, the appearance of a linear trend could be justified as a consequence of general urbanization (e.g. Zagreb-Maksimir and Karlovac), while in some other cases it

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could have been caused by the planting or uprooting of trees in their vicinity (e.g. a phenological garden was planted in Križevci). No homogeneity break has been discovered at three weather stations in the mountainous area (Ogulin, Gospić and Zavižan) and at two coastal stations (Mali Lošinj and Hvar).

Thus, even in rather complex environmental conditions, good quality observation and a solid metabase (i.e. data such as relocation dates, replacement of instruments, change of environment, etc.) for the time series of annual mean air temperature, sound information can be extracted after the application of the homogenization procedure. The influence of inhomogeneities on the diagnostic parameters of the air temperature time series for Croatia (natural trend indices, correlation functions, etc.) will be considered in Section 5.

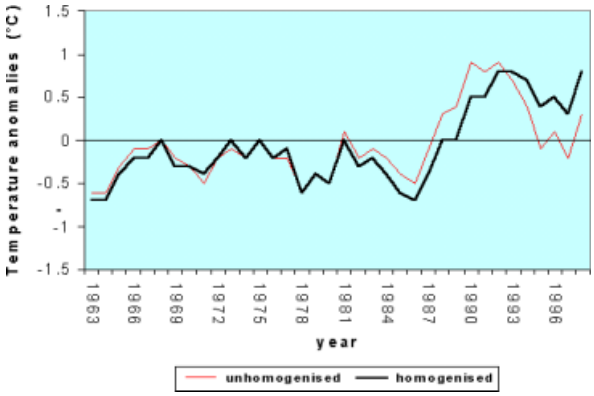



Figure 8. Comparison of the 5-year moving averages of the anomalies of unhomogenised and homogenised time series of mean annual air temperature defined for the reference period 1961–2000, for the Karlovac weather station. This figure is available in colour online at www.interscience.wiley.com/ijoc

## 5. Influence of data inhomogeneity on time-spatial diagnostic parameters

In the introduction, it was mentioned about the possible influence of data inhomogeneity on statistical parameters such as means, variance, extremes, correlation, probability density functions, etc. As it is obvious from the average change examples represented in Table II (the highest being up to 0.5 °C), these changes are difficult to notice compared with the real air temperature time series before and after homogenization because these changes are small when they are compared with the natural variations of mean annual air temperature (in the order of several degrees Celsius). Therefore, a more illustrative representation of these differences is necessary.

One possibility is to consider the 5-year moving average anomalies calculated for homogenized and unhomogenized time series compared to the multiannual averages before and after homogenization, respectively. Indeed, after the application of this procedure, the difference in values before and after homogenization is clear, as the example of the Karlovac weather station shows (Figure 8). Thus, it is obvious that the 5-year moving average anomalies are lower during a trend period while they are higher in average about 0.5 °C after the station has been relocated out of town.

The representation of the 5-year moving averages of unhomogenized (Figure 9(a)) and homogenized (Figure 9(b)) anomalies of the mean annual air temperature time series from 10 weather stations in Northern (lowland) Croatia is also illustrative. A wider range of anomalies is obvious for the unhomogenized series compared with the homogenized ones, which means that the correlation coefficients go mostly up after homogenization. This is confirmed by Figure 10, where the correlation coefficients (between Karlovac and other weather stations) are presented both before and after homogenization. As it was expected, the correlation coefficients are mostly from 0.1 to 0.2 higher for the homogenized smoothed time series than for the unhomogenized (see Section 3).

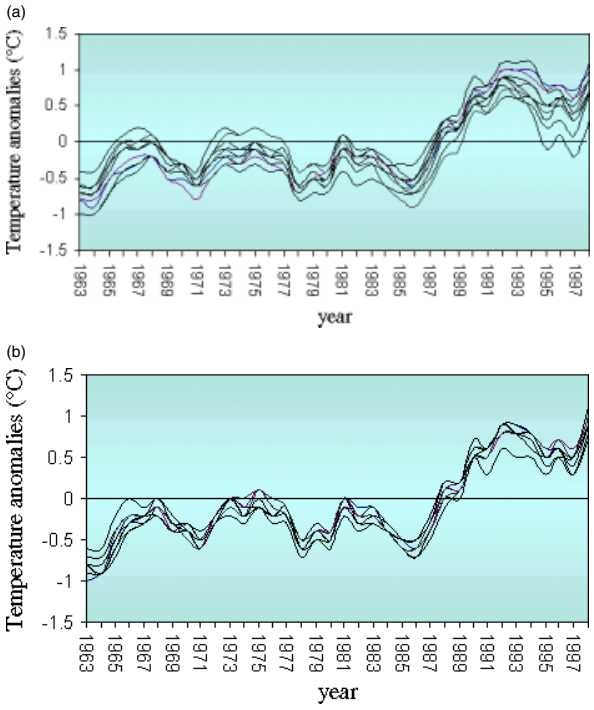



Figure 9. Comparison of the 5-year moving averages of the anomalies of unhomogenised (a) and homogenised (b) of the mean annual air temperature series defined for the reference period 1961–2000, for 10 weather stations in Northern Croatia. This figure is available in colour online at www.interscience.wiley.com/ijoc

All this can be applied to linear trend changes as well. For the calculation of linear trends a simple linear regression equation has been used where discrete time (time step of 1 year) has been taken as an independent and the average annual temperature as a dependent stochastic variable (see e.g. Penzar and Penzar, 2000; Wilks, 2006). The differences between the total linear trends for the homogenized and unhomogenized 22 average annual air temperature time series have been calculated. The results are represented in Figure 11(a). The values range from −0.34 °C/40 years for Zadar to 0.39 °C/40 years for

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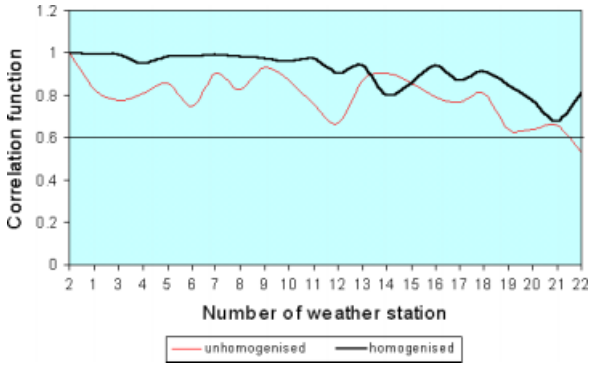



Figure 10. Comparison of the correlation functions for the 5-year moving averages of unhomogenised and homogenised mean annual air temperature series between the Karlovac weather station, designated as number 2 in Figure 1, and other stations, also designated by numbers in the same figure. Higher numbers roughly correspond to higher distance of the other weather stations from Karlovac. This figure is available in colour online at www.interscience.wiley.com/ijoc

Osijek. The 'true' annual temperature linear trends are represented in Figure 11(b). The gradual rise of the linear trend from the south to the north of the Croatian territory in the period considered is very probably a consequence of global warming. Without homogenization a rather false picture would be obtained (see e.g. Vincent and Gullett, 1999).

Finally, the amplitude of the first principal component (PC) for the 5-year mean annual temperature moving averages before and after homogenization has been considered (see also Begert *et al*., 2005; Pandžić, 1986; Preisendorfer, 1988). As it is obvious form Table III, the first PC describes 84% of the total field variance before homogenization and 90% after homogenization. Thus, the statistical structure is more compact (the correlation coefficients are higher) after homogenization. On the other hand, there is no significant difference between the time variation of the amplitude of the PC before and after homogenization, which is represented in Figure 12. This result is in accordance with that of Begert *et al*. (2005) as well as with those of Moberg and Alexandersson (1997). They have stated that the spatial average of mean annual air temperature is 'immune' to the existing inhomogeneities which appear in the mean annual air temperature time series considered for Sweden and Switzerland.

Thus, spatial averaging acts as a kind of filter for the rejection of inhomogeneities appearing in a particular time series. A similar conclusion can be applied even to a global scale (Easterling and Peterson, 1995a, b), which could be very important from the point of view of global mean air temperature calculations. According to this, the global air temperature mean could be insensitive to at least one category of the possible inhomogeneities of a particular time series produced in operational meteorological practice. On the other hand, this immunity can be used to define the reference series as a weighted average of the time series of the weather stations surrounding the candidate station (it seems that the use of time series for 3 to 5 surrounding weather stations is reasonable in Croatian conditions).

On the other hand, as it has been shown in the example of the 5-year moving average application, temporal smoothing of the time series emphasizes their inhomogeneities. Therefore, when calculating correlation coefficients or performing a trend analysis, special attention has to be devoted to the homogeneity of the time series. As expected (although in most cases weak in this study)

Table III. Eigen values (%) of the first four principal components (PCs) for the 5-year moving averages of air temperature in Croatia, before (a) and after (b) homogenisation.

| | PC1 | PC2 | PC3 | PC4 |
|---|---|---|---|---|
| a | 84 | 7 | 3 | 2 |
| b | 90 | 7 | 1 | 1 |

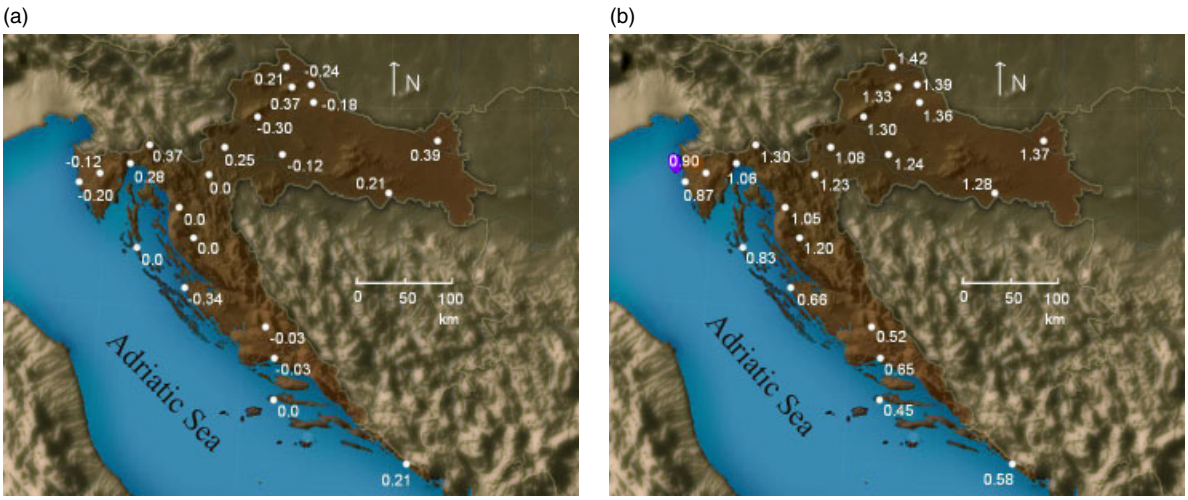



Figure 11. (a) Differences between the total linear trends for the homogenized and unhomogenized 22 average annual air temperature time series for Croatia, in °C/40 years; (b). Linear trends for the homogenized 22 average annual air temperature time series for Croatia, in °C/40 years. This figure is available in colour online at www.interscience.wiley.com/ijoc

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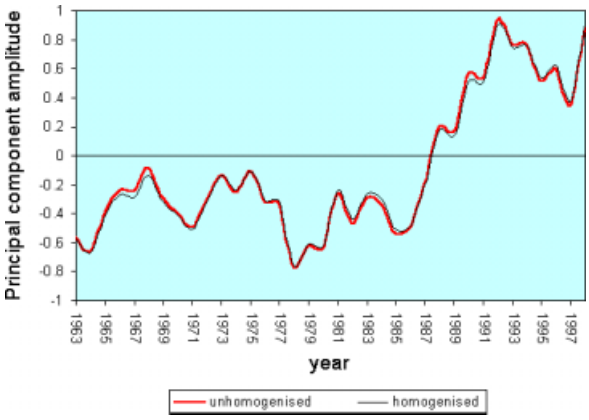



Figure 12. Comparison of the first principal component (PC) amplitudes for the 5-year mean annual temperature moving average anomalies, defined for the period 1961–2000, before and after homogenization. The anomalies refer to the 22 weather stations in Croatia which are shown in Figure 1. This figure is available in colour online at www.interscience.wiley.com/ijoc

the inhomogeneity of the air temperature series has an influence on multiannual averages (i.e. climate normals). Relatively small changes in multiannual averages are, frequently, also a consequence of the variability of the change sign within a particular time series (see Table II). Exceptions can be homogeneity breaks which are consequences of station relocation (e.g. Karlovac) or urbanization (e.g. Zagreb-Maksimir).

The results described refer to mean annual air temperature time series. However, testing seasonal, monthly, daily or extreme values of air temperature can give, in some cases, results which can be different from those obtained for the annual time scale (see e.g. Zwiers and Storch, 1995; Wijngaard *et al*., 2003).

## 6. Conclusion

The SNHT was applied for the detection of inhomogeneities in the time series of mean annual air temperature from 22 weather stations in Croatia for the period 1961–2000. For three weather stations in the mountainous part of Croatia and for two on the Adriatic islands no break has been detected. These are main weather stations of the first quality order, which are exposed to weak changes of environmental conditions and they have not been relocated in the period considered. For six stations, abrupt breaks of homogeneity have been detected (the highest magnitude for Karlovac). For several stations, in spite of their relocation, breaks corresponding to the time of relocation have not been established. Linear trends have been detected at about 10 stations, at some stations multiple ones. In some cases, the explanation for the trend is obvious, e.g. at Zagreb-Maksimir (urbanization) or Križevci (starting a phenological garden) while in some cases the trends are difficult to explain (e.g. for Bjelovar, Durdevac, Parg, Pazin and Rovinj). It is obvious that these are a group of lower-rank stations which resulted in lower quality of observations. In addition to the influence on the multiannual average, the highest of 0.5 °C being at Karlovac (in our case), a comparison of the 5-year moving averages of temperature anomalies indicates the importance of taking into consideration the homogeneity breaks during the study of trends or correlation functions. On the other hand, it has been shown by means of the first PCs for the 5-year moving averages of mean annual air temperature, describing 84% (before homogenization) and 90% (after homogenization) of total field variance and well representing spatial averages for Croatia, that there is no significant difference between PC amplitudes for the corresponding temperature anomalies before and after homogenization. That result is in accordance with those obtained for Sweden and Switzerland as well as on a global scale in some cases. This result is also suitable for the creation of reference time series as averages from 3 to 5 weather stations surrounding the candidate station. The next step is to possibly check the results obtained by independent testing methods although the SNHT can be considered to be a double test. In addition, a higher number of existing weather stations (more than a hundred in Croatia) should be included in the study, the time series should be prolonged in the past and seasonal, monthly and daily time scales should be considered, including extremes.

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## References

Aguilar E, Auer I, Brunet M, Peterson TC, Wieringa J. 2003. *Guidelines on Climate Metadata and Homogenization*. Report WMO-TD 1186, World meteorological Organization: Geneva, Switzerland; 50.

Alexandersson H. 1986. A homogeneity test applied to precipitation data. *Journal of Climatology* **6**: 661–675.

Alexandersson H, Moberg A. 1997. Homogenization of Swedish temperature data. Part I: Homogeneity test for linear trends. *International Journal of Climatology* **17**: 25–34.

Begert M, Schlegel T, Kirchhofer W. 2005. Homogeneous temperature and precipitation series of Switzerland from 1864 to 2000. *International Journal of Climatology* **25**: 65–80.

Brunetti M, Maugeri M, Monti F, Nannia T. 2006. Temperature and precipitation variability in Italy in the last two centuries from homogenised instrumental time series. *International Journal of Climatology* **26**: 345–381.

DeGaetano AT. 1996. Recent trends in maximum and minimum temperature threshold exceedences in the NorthEastern United States. *Journal of Climate* **9**: 1646–1660.

DeGaetano AT. 2006. Attributes of several methods for detecting discontinuities in mean temperature series. *Journal of Climate* **19**: 838–853.

Drogue G, Mestre O, Hoffman L, Iffly JF, Pfister L. 2005. Recent warming in a small region with semi-oceanic climate, 1949–1998: what is the ground truth?. *Theoretical and Applied Climatology* **81**: 1–10.

Ducré-Robitaille JF, Boulet G, Vincent LA. 2003. Comparison of techniques for detection of discontinuities in temperature series. *International Journal of Climatology* **23**: 1087–2003.

Easterling DR, Peterson TC. 1995a. A new method for detecting and adjusting for undocumented discontinuities in climatological time series. *International Journal of Climatology* **18**: 1493–1517.

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Easterling DR, Peterson TC. 1995b. The effect of artificial discontinuities on recent trends in minimum and maximum temperatures. *Atmospheric Research* **37**: 19–26.

Hawkins DM. 1977. Testing a sequence of observations for a shift in location. *Journal of the American Statistical Association* **73**: 180–185.

Herzog J, Müller-Westermeier G. 1996. Homogenization of various climatological parameters in the German weather service. In *Proceedings of the First Seminar for Homogenization of Surface Climatological Data*. Budapest, Hungary, 6–12 October, 1996.

Khaliq MN, Ouarda TBMJ. 2007. A note on the critical values of the Standard Normal Homogeneity Test (SNHT). *International Journal of Climatology* **27**: 681–687.

Li QX, Liu XN, Zhang HZ, Peterson TC, Easterling DR. 2004. Detecting and adjusting temporal inhomogeneity in Chinese mean surface air temperature data. *Advances in Atmospheric Sciences* **21**: 260–268.

Likso T. 2003. Inhomogeneities in temperature time series in Croatia. *Hrvatski meteorološki časopis* **38**: 3–9.

Lindgren BW. 1976. *Statistical Theory* (3rd ed.). Macmillan: London, 614.

Marković RD. 1975. *Mathematical–Statistical Methods of Testing Consistency and Homogeneity of Meteorological and Hydrological Elements of Human Environment.*Federal Hydrometeorological Institute: Belgrade, 48.

Moberg A, Alexandersson H. 1997. Homogeneization of Swedish temperature data. Part II: Homogenized gridded air temperature compared with the subset of global gridded air temperature since 1861. *International Journal of Climatology* **17**: 35–54.

Mood AM, Graybill FA, Boes DC. 1974. *Introduction to the Theory of Statistics.*McGraw-Hill: Kogakusha, Tokyo, 564.

Pandžić K. 1986. Factor analysis of temperature field on a relatively small area. *Idojaras* **90**: 321–331.

Pavlić I. 1970. *Statistical Theory and Application (in Croatian)*. TehniŠka knjiga: Zagreb, 343.

Penzar I, Penzar B. 2000. *Agrometeorology (in Croatian)*. Školska knjiga: Zagreb, 222.

Peterson TC, Easterling DR. 1994. Creation of homogeneous composite climatological reference series. *International Journal of Climatology* **14**: 671–679.

Peterson TC, Easterling DR, Karl TR, Groisman P, Nicholls N, Plummer N, Torok S, Auer I, Boehm R, Gullett D, Vincent L, Heino R, Tuomenvirta H, Mestre O, Szentimrey T, Salinger J, Forland EJ, Hanssen-Bauer I, Alexandersson H, Jones P, Parker D. 1998. Homogeneity adjustments of in situ atmospheric climate data: a review. *International Journal of Climatology* **18**: 1493–1517.

Preisendorfer RW. 1988. *Principal Component Analysis in Meteorology and Oceanography.*Elsevier: Amsterdam, 425.

Reeves J, Chen J, Wang XL, Lund R, Lu Q. 2007. A review and comparison of changepoint detection techniques for climate data. *Journal of Applied Meteorology and Climatology* **46**: 900–915.

Steffensen P. 1996. *Standard Normal Homogeneity Test for Windows*. DMI Technical Report No. 9613, Copenhagen, 35.

Tuomenvirta H, Alexandersson H. 1996. Review on the methodology of the standard normal homogeneity test. In *Proceedings of the First Seminar for Homogenization of Surface Climatological Data* Budapest, Hungary, 6–12 October, 1996.

Vincent LA. 1998. A technique for the identification of inhomogeneities in Canadian temperature series. *Journal of Climate* **11**: 1094–1104.

Vincent LA, Gullett DW. 1999. Canadian historical and homogeneous temperature datasets for climate change analyses. *International Journal of Climatology* **19**: 1375–1388.

Wilks DS. 2006. *Statistical Methods in the Atmospheric Sciences.*Elsevier: Amsterdam, 627.

Wijngaard JB, Klein Tank AMG, Konnen GP. 2003. Homogeneity of 20th century European daily temperature and precipitation series. *International Journal of Climatology* **23**: 679–692.

Zwiers FW, von Storch H. 1995. Taking serial correlation into account in tests of the mean. *Journal of Climate* **8**: 336–351.

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